

All-Dielectric Heterogeneous Metasurface as an Efficient Ultra-Broadband Reflector

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Conventional all-dielectric metasurfaces made from high-index and low-loss materials (such as Si or GaAs) can achieve high optical efficiencies as well as broadband characteristics, mostly for the infrared (IR) light. However, maintaining the bandwidth and efficiency for conventional all-dielectric metasurfaces for visible or near-IR light becomes challenging, due to the limitations from material selection as well as fabrication techniques. In this paper, the idea of heterogeneous all-dielectric metasurfaces is proposed for a better metasurface design in shorter-wavelength ranges. An ultra-broadband reflector, as a proof-of-principle demonstration, is presented in this paper, both numerically and experimentally. The broadband reflector demonstrates a good optical efficiency, a broadband reflectance, as well as a great manufacturability. Nanoimprint lithography patterning process and advanced etching recipes are developed for the large-area and low-cost heterogeneous metasurface fabrication.

1. Introduction

In recent years, the breakthroughs in photonics studies^[1] and nanofabrication technologies^[2,3] have enlightened many designs and implementations of optical metasurfaces,^[4–7] where light can be effectively manipulated by the nanoscale patterns built on thin-film materials. In order to achieve the desired functionality, materials in optical metasurfaces are usually expected to have a strong interaction with light.

The conventional coupling mechanisms between light and materials in optical metasurfaces can be predominantly classified into two types, the photon–plasmon coupling^[8] and the photon–phonon coupling.^[9] In the first type, the interaction usually takes place in certain metals such as Au or Ag, where the energy and momentum exchange happens between the incident photons and the free electrons in metals. In the photon–phonon coupling mechanism, the incident photons interact with acoustic photon vibrations in compounds such as SiC.^[10,11] Based on the strong light–matter coupling mechanisms, optical metasurfaces with well-designed structures have aroused numerous applications, such as spectrum filters,^[12,13]

polarization converters,^[14] negative-index metamaterials,^[15,16] vortex generator,^[17] beam detector,^[18] etc.

The common drawbacks in many metasurfaces are their limited operating wavelength bandwidths and their low optical efficiencies, which are largely attributed to their operating mechanisms. The coupling between photon and plasmon (or photon) is not a broadband mechanism and is only effective within a limited wavelength range.^[8] Moreover, as the photons exchange energy or momentum with plasmons or phono-polaritons, a substantial amount of optical energy gets dissipated, due to the low coupling efficiency or material absorption. The limitations of operating bandwidth and optical efficiency could be partially mitigated, to some extent, by optimizing the designs of

metasurfaces,^[19,20] but they cannot be completely surmounted because of the fundamental properties of materials and mechanisms of light–material interaction.

Recently, metasurfaces that are solely built on dielectrics^[21] have started to draw more and more attentions for a broader bandwidth as well as for a higher efficiency. The light interaction mechanism in all-dielectric metasurfaces differs fundamentally from those in conventional metasurfaces. In all-dielectric metasurfaces, the incident light interacts with dielectric materials in which the nanoscale patterns have been defined to achieve the desired optical resonances for certain functionality. To enhance the coupling between light and dielectrics, the refractive index of the materials should be high. Meanwhile, to achieve a higher optical efficiency, the dielectric loss should be low in the operating wavelength range. Therefore, Si or GaAs, for example, are good candidates for building optical metasurfaces in infrared (IR), for their high refractive index and low optical loss. Recently, all-dielectric metasurfaces have been demonstrated as high-reflectivity broadband spectral filters,^[22] flat lenses,^[23] zero-index^[24] or anisotropic^[25] metamaterials, etc.

However, developing all-dielectric metasurfaces becomes more challenging for shorter wavelengths, in the near-IR or visible range. In those wavelength ranges, more potential practical applications of metasurfaces can be potentially proposed, since human eyes can directly sense and respond to light there. The first challenge comes from the fact that the dielectrics with both a high index and a low loss at shorter wavelengths become more difficult to find. For example, Si and GaAs are no longer good choices in visible range because of the strong absorption.

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Fundamentally, it becomes physically improbable for materials to have both high refractive index and low optical loss (i.e., large bandgap energy) at shorter wavelengths, due to the fundamental relation between the material index and bandgap energy.^[26] For this material limitation, compromises either on the refractive index or the material loss have to be taken, so the performance of all-dielectric metasurfaces for shorter wavelengths will be adversely impacted.

The second challenge on realizing all-dielectric metasurfaces for shorter wavelengths comes from the fabrication techniques. As the targeting wavelengths get smaller, the structural dimensions on metasurfaces should be accordingly scaled down to avoid unwanted higher-order diffractions. Photolithography^[27] or electron beam lithography^[2] becomes either incapable or dawdling for large-area metasurfaces demonstration. Basically, all-dielectric metasurfaces in shorter wavelength ranges require high-resolution, high-throughput, and low-cost fabrication technologies.

In this paper, the design paradigm of heterogeneous all-dielectric metasurfaces is proposed to cope with the above limitations for all-dielectric metasurfaces. In heterogeneous all-dielectric metasurfaces, the material issues that restrained many conventional all-dielectric metasurfaces can be effectively circumvented by incorporating different types of dielectric materials. In heterogeneous all-dielectric metasurfaces, materials with a high loss and high refractive index are integrated with materials with a low loss and low refractive index. Compared to materials with both high index and low optical loss, which do not exist in nature, those two types of materials are more available for shorter wavelengths in visible or near-IR range. High-index materials serve to enhance the optical coupling between incident light and metasurfaces structures while low-index materials function as a low-loss carrier for optical modes. In this way, optical metasurfaces can be designed with both a wide bandwidth and a high optical efficiency for shorter wavelengths, where it would be challenging for many conventional all-dielectric metasurfaces.

As a proof-of-principle demonstration of the design paradigm, an ultra-broadband reflector is both numerically and experimentally demonstrated for high reflectivity in both visible and near-IR wavelengths. We should clearly point out that many interesting work on optical perfect reflectors have been demonstrated previously. For example, broadband reflector can be conventionally achieved with planar multilayer dielectrics, such as distributed Bragg reflectors (DBRs), but the bulkiness of the device and the high cost of multilayer dielectric deposition greatly limit its application. In comparison, nanostructured devices can be more compact and versatile. In IR range, 2D array of Si particles has been widely used as the building block to achieve perfect reflectors,^[28–31] but the inclusion of bulky silicon blocks makes those designs challenging to work for shorter wavelengths while maintaining a similar optical efficiency. In this paper, the operating wavelength range of the ultra-broadband reflector, with the heterogeneous all-dielectric design, can be extended from IR to visible with a minimized efficiency loss. For metasurface patterning, a large-area, low-cost, and high-resolution nanoimprint^[32] process is developed as the lithographical tool. In addition, advanced etching recipes are developed to etch the heterogeneous metasurfaces with the

desired profile. The reflection bandwidth of the heterogeneous ultra-broadband reflector is greatly improved by high-index (but high-loss in visible or near-IR range) materials and a good optical efficiency can still be maintained by optimizing the heterogeneous design. In addition, the good reflectivity in the all-dielectric metasurface is very tolerable to structural parameter variations and enjoys a relaxed tuning range, so it can be widely used as the basic building blocks for high-efficiency and broadband metasurface devices for more potential applications. Moreover, all materials included in the heterogeneous metasurface design are being used in standard integrated circuit (IC) fabs, so those metasurfaces can be massively produced using the existing semiconductor infrastructure.

2. Theoretical and Numerical Results

2.1. Heterogeneous Metasurface Design

The design philosophy behind the heterogeneous all-dielectric metasurfaces is that incorporating lossy dielectrics will not necessarily kill the overall optical efficiency. The energy loss of an electromagnetic system is proportional to

$$\iiint \varepsilon''(x, y, z) |\vec{E}(x, y, z)|^2 dx dy dz \quad (1)$$

In Equation (1), $\varepsilon''(x, y, z)$ denotes the imaginary part of the local electric permittivity at the coordinates (x, y, z) and it is a measure of optical absorption. From Equation (1), one can see that the overall electric energy dissipation is not only dependent on the material loss, but also on the electric field distribution. If the electric field distribution (i.e., optical resonances) can be manipulated by the design of metasurfaces so that there is a minimized overlap between optical modes and lossy materials in the system, a good overall optical efficiency can still be achieved. Taking the optical cavity as an example (Figure 1), it has metallic mirrors on both ends but it can still reach very high Q (quality factor) because the metallic (lossy) regions have very little overlap with the optical fields where the reflected and incident light interfere destructively. Similarly, for heterogeneous all-dielectric metasurfaces, the impact of lossy materials on the overall optical efficiency can be effectively mitigated by optimizing the metasurface topology so that the optical resonance distribution gets manipulated for both the desired functionality and optical efficiency. Compared to conventional

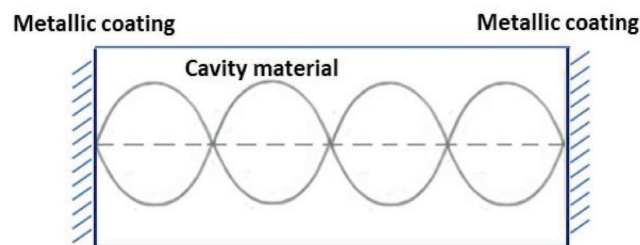


Figure 1. The conceptual diagram of an optical cavity that shows overall optical energy loss can be low even though there are lossy regions.

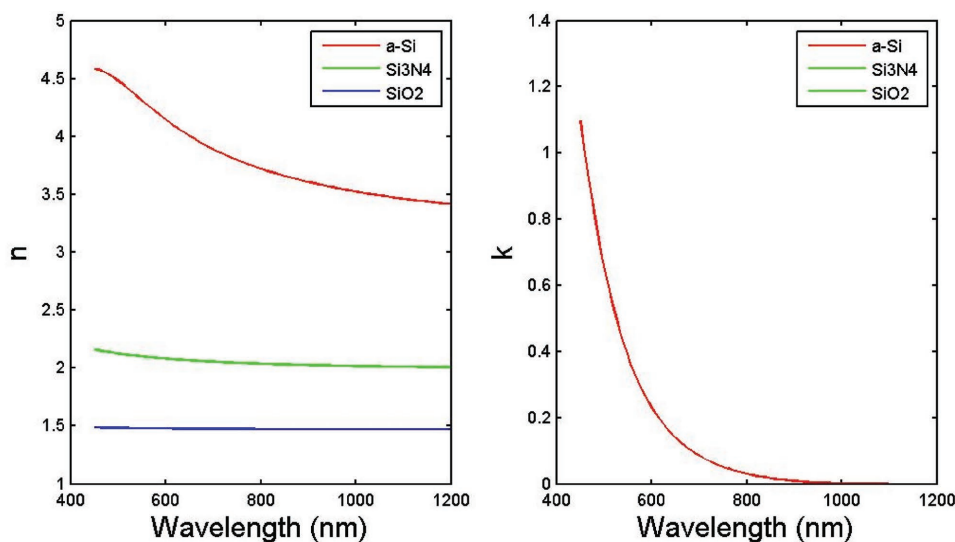


Figure 2. Measured refractive index (n, k) of materials include in the heterogeneous all-dielectric design.

all-dielectric metasurfaces, heterogeneous all-dielectric metasurfaces will have a greater freedom in design and optimization.

In the heterogeneous all-dielectric metasurface design demonstrated in this paper, three materials, amorphous silicon (a-Si), silicon nitride (Si_3N_4), and silicon dioxide (SiO_2), are selected to build the optical metasurfaces in visible and near-IR range. Of all three types of materials here, a-Si has the highest index and the highest absorption, while Si_3N_4 and SiO_2 have lower index and lower absorption (**Figure 2**). They are very available and low-cost materials and their range of index contrast are high enough to enable applications in visible or near IR range. The role of a-Si is to enhance the light–structure coupling while the role of Si_3N_4 and SiO_2 is to sustain the optical resonance with little absorption. By optimizing the structures and parameters, the impact of a-Si absorption can be minimized so that both strong light coupling and high optical efficiency can be simultaneously achieved.

2.2. Broadband Reflector Optimization

The goal of the heterogeneous all-dielectric metasurface design is to achieve an ultra-broadband reflector, not in infrared range

where Si or GaAs can be directly applied as in conventional all-dielectric metasurfaces, but in shorter wavelengths where both design and fabrication become more challenging. In IR range, metasurfaces with a broadband reflectivity has been achieved,^[28–31,33] using gratings or arrays of Si particles. However, it would be difficult to scale down the same design to shorter wavelengths, since Si or GaAs cannot be used in the same way as we did in IR range. Alternative materials such as TiO_2 have been reported to replace Si for visible range applications,^[34–36] but the reflection characteristics are not as good due to its lower refractive index, compared to that of Si or GaAs in IR range.

In the design of the heterogeneous all-dielectric metasurface (hybrid grating), the high index material, a-Si, will be incorporated for a stronger interaction between the light and the metasurface to achieve the broadband reflectivity for shorter wavelengths. Meanwhile, the addition of a-Si must be controlled to avoid the optical efficiency degradation. Si_3N_4 and SiO_2 , due to their little optical absorption, will be used to hold the optical resonances, while a-Si will be used to enhance the local confinement of optical resonances. Therefore, it is most sensible to place the thin crust of a-Si at the boundary between ambient environment and the metasurface, in order to enhance light–matter interaction as well as to maintain a good optical

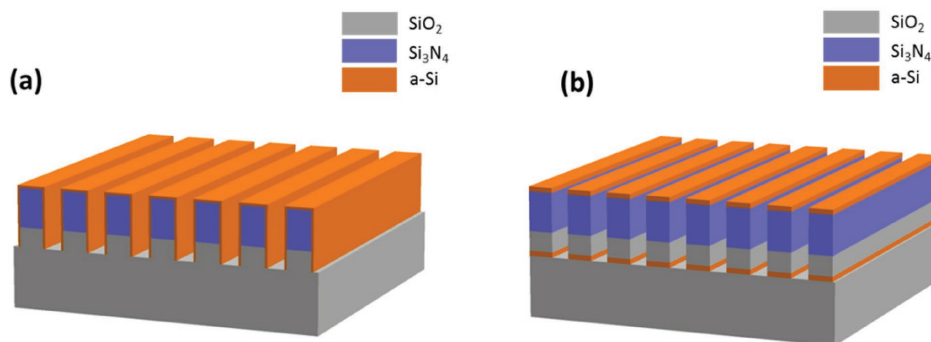


Figure 3. a) Schematic of the core-shell structure. b) Schematic of the multilayer structure.

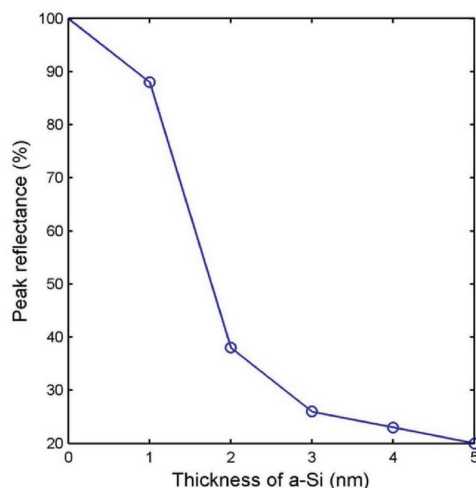


Figure 4. Optical efficiency degradation in the optimized core-shell structure with different a-Si thicknesses.

efficiency. To elucidate this point, two designs were investigated, which are shown in **Figure 3**, the core-shell structure and the multilayer structure. In the core-shell structure, the crust of a-Si forms a conformal coverage on the grating blocks that are composed of Si_3N_4 and SiO_2 . In comparison, the multilayer structure has no a-Si coverage on the sidewall of the grating blocks. Incorporating both Si_3N_4 and SiO_2 in each grating block can provide us with an additional freedom of optimization for better optical characteristics.

For the core-shell structure, its optical efficiency degrades heavily with the inclusion of a-Si crusts, even with optimized structures. As shown in **Figure 4**, the reduction in reflection peak indicates the degradation of optical efficiency due to the absorption of a-Si. With a 5 nm thick a-Si crust, the reflection peak is only 20%, which makes this structure an unwise choice. The fundamental reason is that the optical mode is strong on the sidewall of the grating blocks, and once the strong resonances overlap a lot with the lossy a-Si crust on the sidewall, the overall optical efficiency will be reduced. By optimizing parameters alone, the nature of optical resonances and its overlap with the a-Si crust on the sidewall cannot be effectively minimized.

In the multilayer structure, on the contrary, there are no a-Si crusts on the grating sidewall where the optical resonances are very strong. A thin layer of a-Si is placed both at the top and bottom of each grating block to increase local light confinement. In this way, the material loss of a-Si can get well controlled and a good reflectance can still be achieved with the inclusion of a-Si. As shown in **Figure 3b**, each grating block consists of four layers of dielectrics, a-Si, Si_3N_4 , SiO_2 , and a-Si (from top to bottom). The top and bottom a-Si layers serve to enhance the contrast between the metasurface and ambient environment so that light can be more effectively coupled and confined locally in the metasurface. Si_3N_4 and SiO_2 layers serve as the low-loss medium to support the optical resonance. The ratio between Si_3N_4 and SiO_2 can help to finely tune the optical mode distribution for even better optical efficiency by providing one more degrees of freedom to engineer the metasurface.

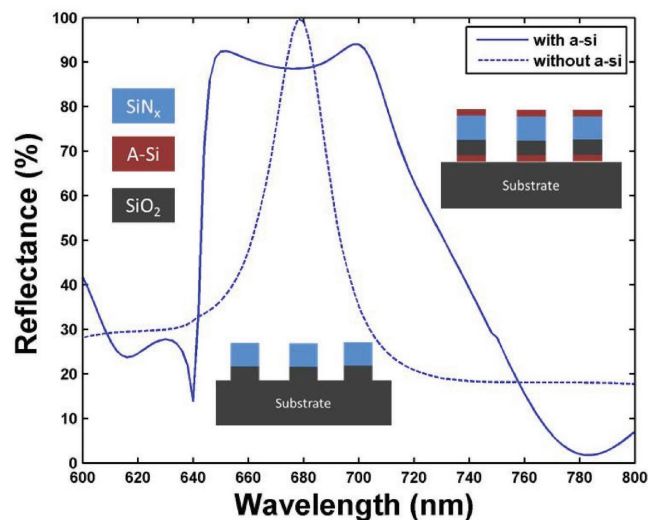


Figure 5. Simulated reflectance (TE-polarized: perpendicular to grating direction) of the optimized heterogeneous all-dielectric metasurface (optimized with and without a-Si layers).

The addition of SiO_2 below Si_3N_4 layer is necessary for the optimal optical performance for it can help to increase the contrast between the Si_3N_4 layer and its adjacent area. On the one hand, this bilayer setup helps to improve optical efficiency by reducing the field overlap with lossy Si layers at the bottom. On the other hand, the guided mode held by this structure will become more leaky (due to the increased contrast in the bilayer structure) which helps to expand the reflection band.^[37]

The reflectance of the multilayer heterogeneous all-dielectric metasurface is shown in **Figure 5**. The pitch of the grating is 455 nm and the line width is 250 nm. The thicknesses of the four layers of dielectrics are 40, 400, 200, and 50 nm (from top to bottom). The optimized parameters lead to a broadband reflectance with an average reflectivity above 90%. The inclusion of a-Si layers is the key to the broadband characteristics. To prove this point, **Figure 5** also shows that if a-Si layers are excluded, the reflectance curve (dashed line) becomes much narrower, even after re-optimization of the structure.

A comparison of Ex (absolute value) distribution (with a polarization direction perpendicular to gratings) is shown in **Figure 6**. Without the a-Si layers, the grating is composed of Si_3N_4 and SiO_2 , which has a similar structure to guided mode gratings.^[38] Due to a less index contrast, the guided modes supported here are less leaky and the reflection band is narrower. With the inclusion of a-Si layers, however, the leaky guided mode can be effectively supported due to the increased contrast from the a-Si layers. To maintain a good optical efficiency, the optical fields must also be optimized to minimize its overlap with the lossy region. Therefore, the heterogeneous structure can demonstrate both broadband reflection and good optical efficiency. As shown in **Figure 7**, the optical loss in the reflection band can be reduced below 10% in the reflection band and most high-loss regions are out of the range that this metasurface is designed for.

All above discussions show how important is the heterogeneous structure to realizing the broadband reflectance in

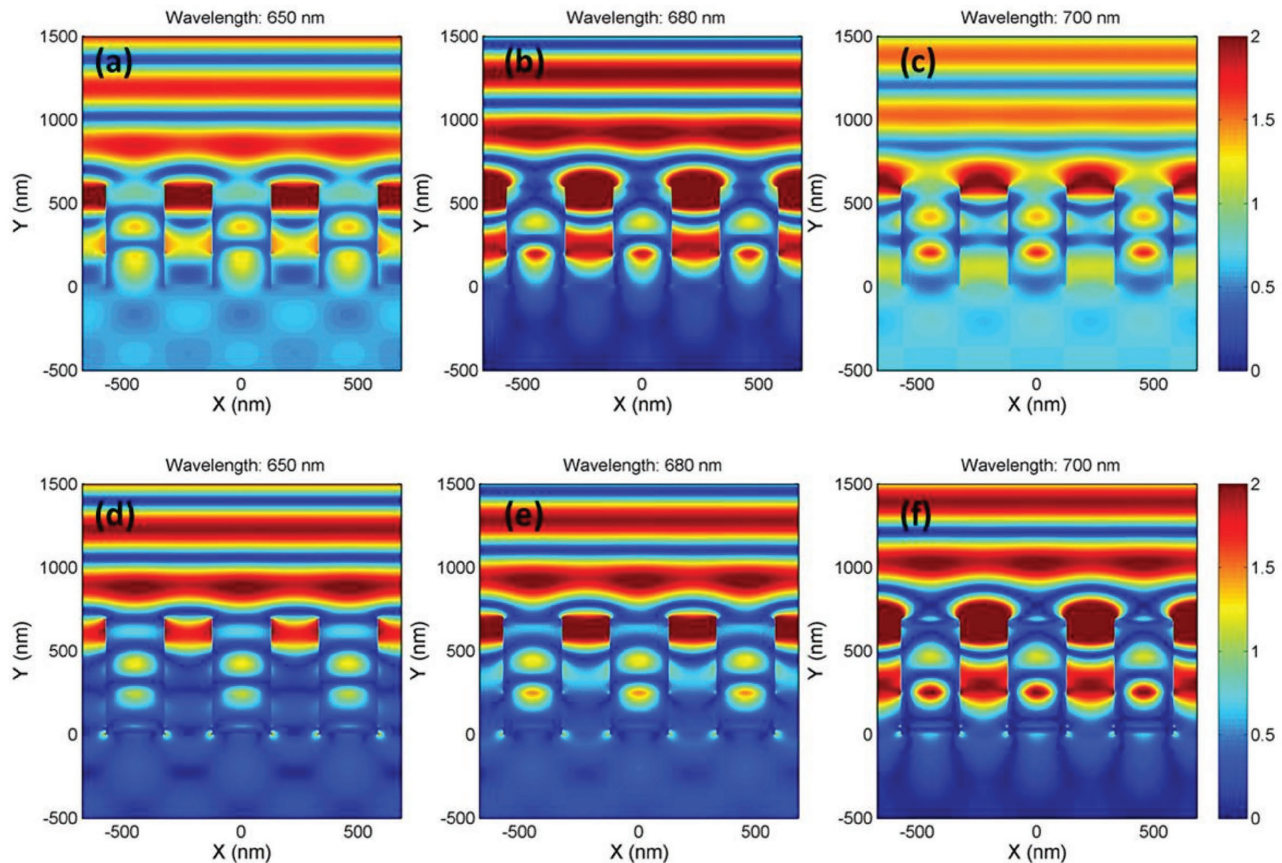


Figure 6. a–c) $|E_x|$ distribution of heterogeneous metasurface without a-Si (wavelength: 650, 680, and 700 nm, respectively). d–f) $|E_x|$ distribution of heterogeneous metasurface with a-Si crust both at top and bottom (wavelength: 650, 680, and 700 nm, respectively). The heterogeneous metasurface has a broader reflection.

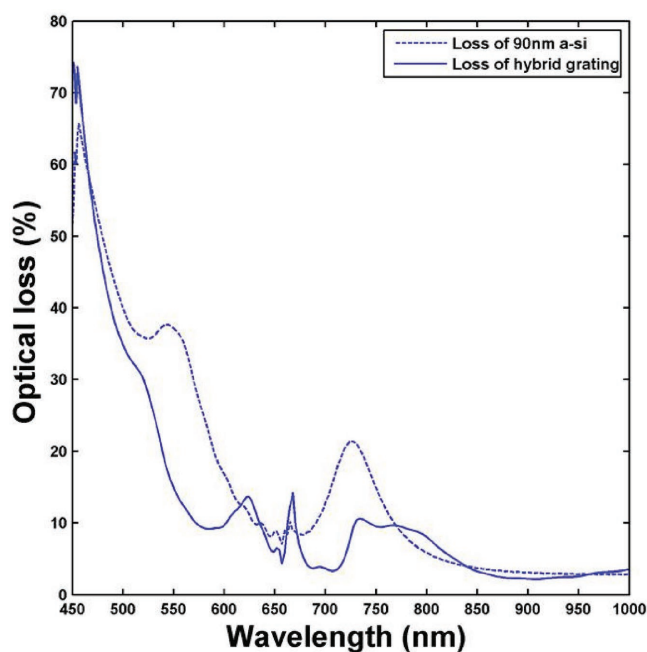


Figure 7. Optical loss (absorption) comparison between a-Si and the heterogeneous metasurface.

shorter wavelength ranges. Achieving a similar performance is very challenging for conventional homogeneous structures, where a single material is used in bulk. A homogeneous Si_3N_4 -based structure can achieve an ideal reflection peak (dashed curves in Figure 5) due to little dielectric absorption, but the reflection bandwidth cannot be broad since the absence of leaky guided mode resonance in low-contrast structures. On the other hand homogeneous a-Si-based structure can achieve a broadband reflectance in IR since a-Si helps to create a high index contrast for leaky guided mode resonance. However, in such structures, good optical efficiency cannot be maintained for short wavelength ranges (640–720 nm) in which a-Si is very absorptive.

2.3. Fabrication Tolerance Analysis

Unlike conventional grating devices, which usually have a strict requirement on structural parameters, the heterogeneous all-dielectric metasurface design is highly tolerable to parameter variations. **Figure 8** demonstrates its tolerability for thicknesses of Si_3N_4 layer and SiO_2 layer. Similar reflectance can be maintained within a big thickness variations (Si_3N_4 : ± 40 nm, SiO_2 : ± 50 nm). We would like to attribute the insensitiveness to Si_3N_4 or SiO_2 to the incorporation of high-index and lossy a-Si, since

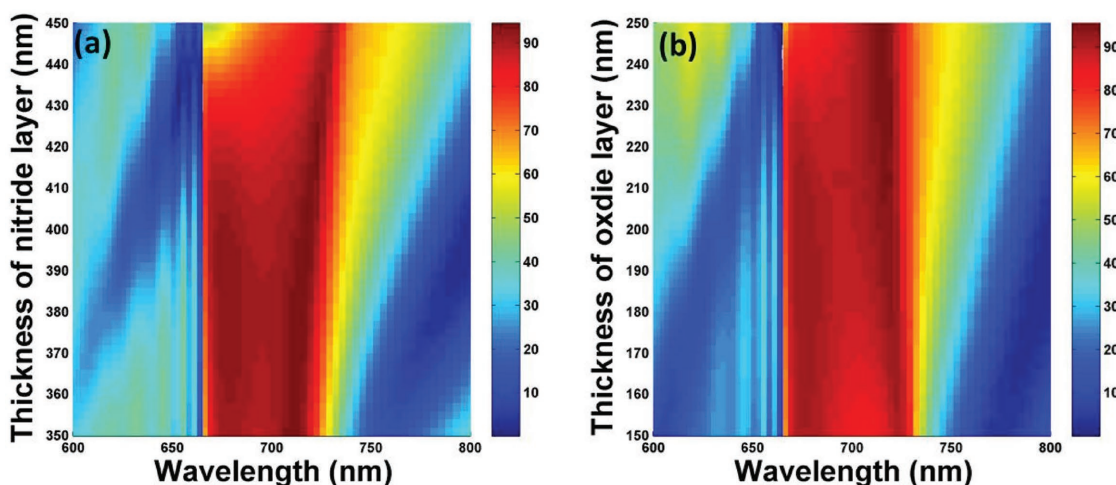


Figure 8. a) Tolerability for Si₃N₄ layer thickness. b) Tolerability for SiO₂ layer thickness.

in guided mode resonance gratings the optical property (e.g., reflection peak) is more sensitive to structural variations. In comparison, its optical property is more sensitive to the thicknesses of a-Si layers. If the layers are too thin, the lack of optical confinement will lead to a narrower bandwidth. If the layers are too thick, the overall optical efficiency will be reduced due to the excessive material loss. Noting that the a-Si thicknesses are small, it will be easier to achieve a consistent thickness in the deposition process. The sidewall angle is a measure of the verticality of the etching profile (Figure 9a) and the design has a good tolerance for sidewall angle when it is below 90° (Figure 9b). As sidewall angle becomes lower, the reflectance bandwidth increases and the center wavelength shifts to the longer wavelength. Finally, the impact of grating line width is shown in Figure 10. Under different line widths, the broadband reflectance can be maintained and the reflection bandwidth can be tuned by different grating line widths. The great structural

tolerance of the heterogeneous metasurface design is a huge advantage over conventional optical metasurfaces.

3. Conclusion

Heterogeneous all-dielectric metasurfaces can effectively overcome the limitations of conventional optical metasurfaces for applications in visible or near-IR range. The inclusion of high-loss and high-index dielectrics provides stronger light-matter interaction and the optical resonances are sustained by low-loss and low-index dielectrics. With an optimized design, the material loss from the lossy dielectrics can be mitigated by manipulating the optical resonances to minimize the overlap with the lossy region in the metasurface. Following the design paradigm of heterogeneous all-dielectric metasurfaces, an ultra-broadband reflector in visible-to-near-IR range is accomplished by using

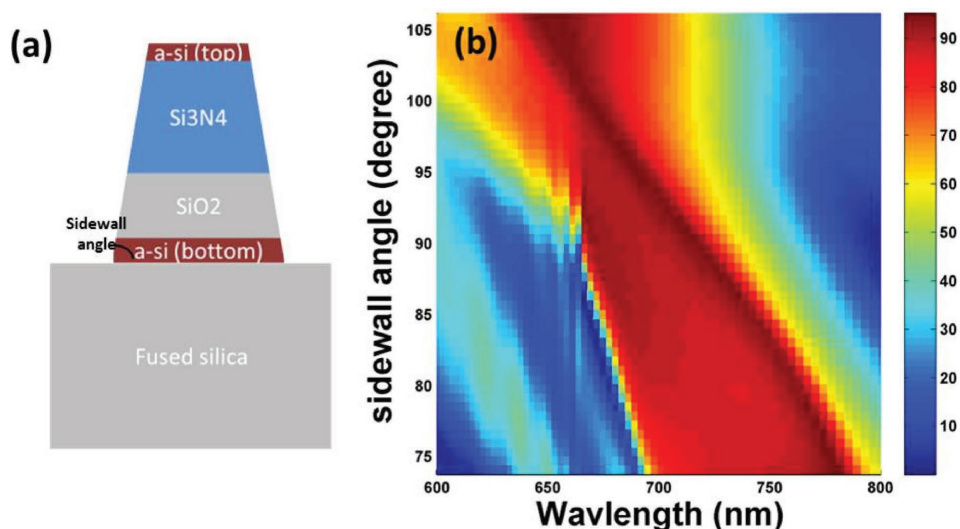


Figure 9. a) Definition of sidewall angle. b) Tolerability for sidewall angle.

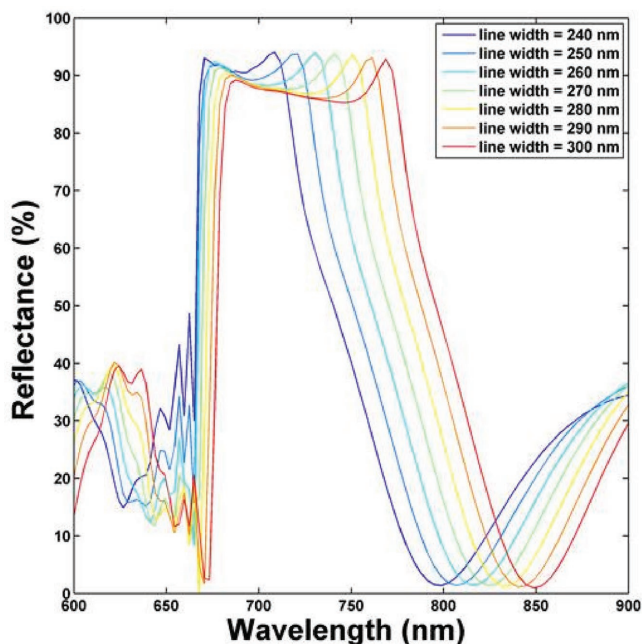


Figure 10. Reflectance curves under different grating line widths.

three types of common dielectrics, SiO_2 , Si_3N_4 , and a-Si. The incorporation of a-Si is optimized so that the reflectance bandwidth gets extended with a minimal compromise on the reflectivity. The heterogeneous design also shows a great tolerance for structural variation, so it has good manufacturability and can be widely used as a building block for more potential applications.

4. Experimental Section

Fabrication Process: To experimentally demonstrate the large-area heterogeneous all-dielectric metasurfaces, nanoimprint lithography

Table 1. Cryogenic etching recipe in metasurface fabrication.

SF_6	O_2	Forward power	ICP power	Temperature	Pressure
30 sccm	20 sccm	150 W	0 W	-100°C	5 mTorr

(NIL) process was specially optimized for patterning the structure, for an accurate control of dimension, and a reduction of edge roughness.^[39] The NIL master mold was fabricated using interference lithography^[40] and a hybrid molding process^[41] was used to transfer, with minimum defects, the pattern from the master mold to the metasurface. After the NIL patterning, the etching recipe at cryogenic temperature was developed to etch the multilayer structure to achieve the desired etching profile.^[42]

The major fabrication steps are depicted in **Figure 11**. First, dielectrics were deposited on SiO_2 substrate with a desired thickness by using plasma enhanced chemical vapor deposition (PECVD) system (Oxford 80+). Then, NIL resist was coated and patterned by nanoimprint lithography using customized setups, where the line width was previously adjusted based on the design. After residual resist etching, Cr evaporation, and lift-off process, Cr mask was formed with the correct pitch and line width.

After Cr mask was formed, reactive ion etching (RIE) recipes were developed to etch the dielectric materials with a good etching profile. The development of etching recipes is very challenging because four layers consisting of three different materials need to be etched with great sidewall passivation. In SiO_2 , compared to Si_3N_4 or a-Si, the Si–O chemical bonding energy is larger than that of Si–N bond or Si–Si bond in a-Si.^[43] Since SiO_2 is harder to etch than Si_3N_4 and a-Si, the protection of Si_3N_4 and a-Si layers that are above SiO_2 is very critical. The typical room-temperature fluorine-based passivant (SiF_x) cannot provide the adequate protection for the softer materials above when etching the harder SiO_2 layer underneath. In order to achieve the desired etching profile for all layers, cryogenic etching (Oxford PlasmaLab 150 Inductively Coupled Fluorine Etcher) recipes were developed. Unlike the conventional fluorine-based passivation mechanism, a different passivant, SiOF_x , was formed at cryogenic temperature. The SiOF_x passivation layers are very protective and it will desorb at room temperature. The formation of SiOF_x can be controlled by the oxygen concentration, plasma power, and etching temperature. The etching recipe is shown in **Table 1** and the cross-sectional scanning electron

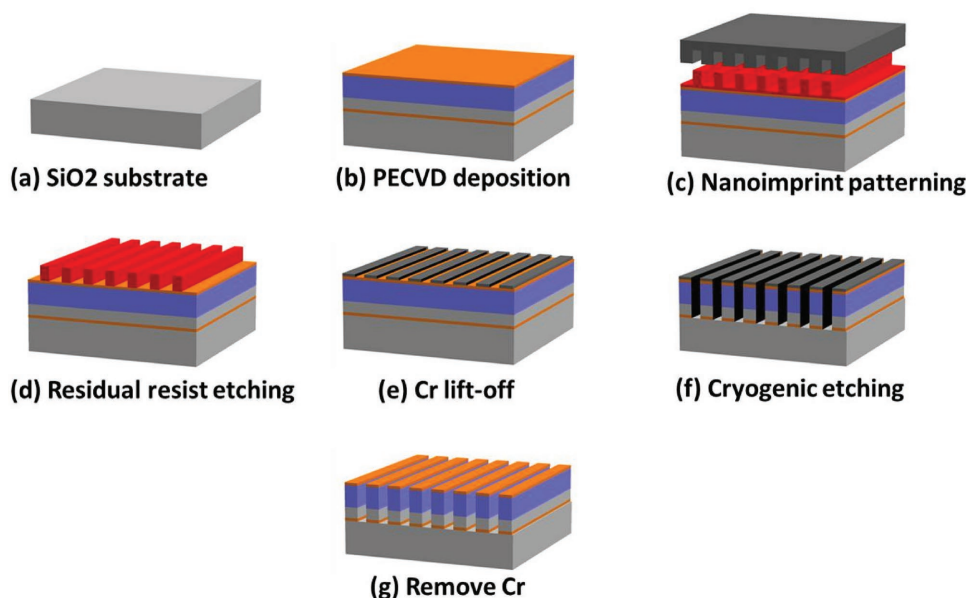


Figure 11. Fabrication process of the heterogeneous all-dielectric metasurface.

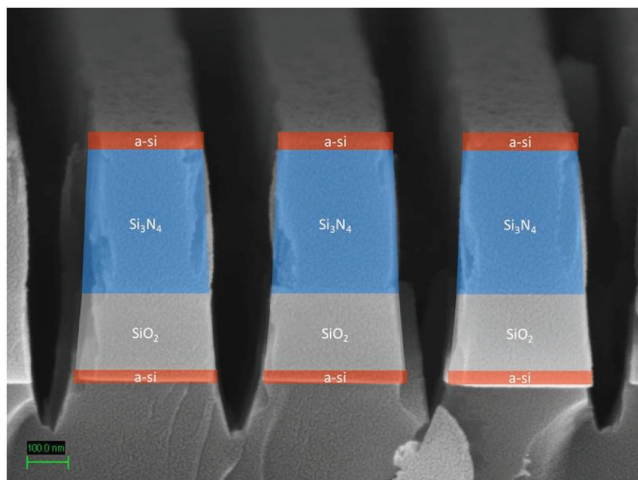


Figure 12. Cross-sectional SEM image of the fabricated metasurface.

microscope (SEM) image of the etching profile is shown in **Figure 12**. The over etching into the substrate is intentional since any residual a-Si will degrade optical efficiency.

Optical Measurement: The reflectance (TE-polarized) of the device was measured and cross-verified using different setups with different spectrometers, Ocean Optics HR-4000 and Lambda 950 UV-vis. The measured reflectance and absorption is shown in **Figure 13**. The fabricated heterogeneous all-dielectric metasurface can provide a good reflection (above 90%) from 600 to 800 nm and the optical absorption above 600 nm is lower than 5% by the heterogeneous metasurface design. Compared with the simulation result in Figure 10, the measured curve shows wider reflection band, due to several reasons. First, the characterization of dielectric index may be inaccurate as the deposited film quality is dependent on the chamber condition (the depositing rate may also vary, but this should be secondary since this device is insensitive to thickness variation). Second, the nonvertical sidewall after cryogenic etching will broaden the reflection band (Figure 9). Finally, the duty cycle of the grating and the over etching into the substrate can also contribute to the wider bandwidth. The inclusion of high-index a-Si effectively broadens the reflection band by enhancing the optical coupling and confinement and the optical field distribution is manipulated by the structural and parametric optimizations to reduce

Table 2. Comparison of metasurfaces with different structures and materials.

Metadevices	Materials	Wavelength range	Normalized bandwidth	Optical efficiency
Guided mode grating ^[38]	HfO ₂ /SiO ₂	840–880 nm	4.6%	>99%
HCG (visible range) ^[36]	TiO ₂ /SiO ₂	500–600 nm	18.1%	>60%
HCG (IR range) ^[33]	Si/SiO ₂	1300–1800 nm	32.2%	>99%
2D array of Si ^[29]	Si/SiO ₂	1550–1800 nm	15.9%	>95%
This work	a-Si/Si ₃ N ₄ /SiO ₂	600–800 nm	28.5%	>90%

the impact of the material loss of a-Si. The comparison of this work over other metadevices is summarized in **Table 2**. The normalized bandwidth (bandwidth normalized to center wavelength) of this work is comparable to that of high contrast gratings (HCG) in IR, while an average reflectance larger than 90% can still be achieved.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

all-dielectric metasurfaces, broadband reflector, cryogenic etching, nanoimprint lithography

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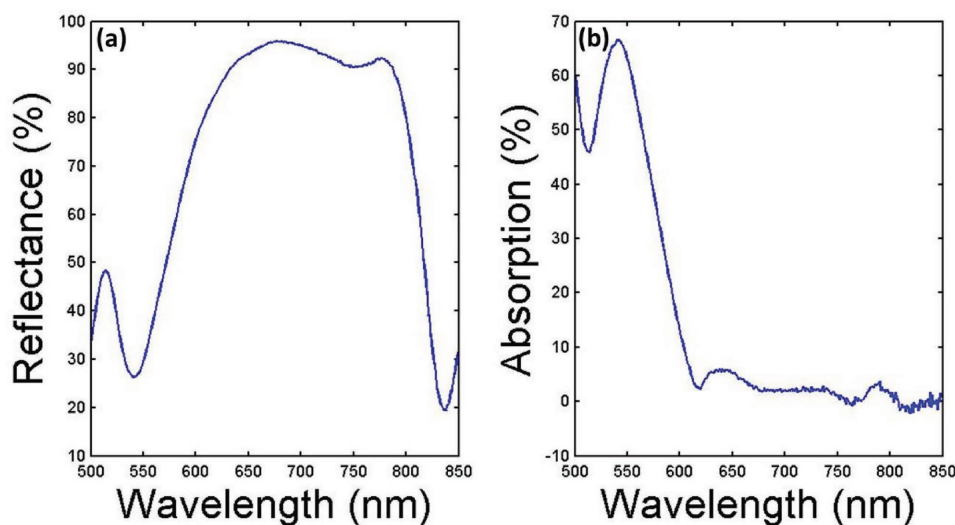


Figure 13. Measured reflectance and absorption (TE-polarized) of the fabricated metasurface.

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